1. If $S=\left\{x \in \mathbb{R}: \sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}\right\}$ then $\sum_{x \in S}\left(\sin\left(\left(x^2+x+5\right)\frac{\pi}{2}\right)-\cos\left(\left(x^2+x+5\right)\pi\right)\right)$ is equal to _________.

[2023 (13 Apr Shift 1)]

2. For $x \in(-1,1]$, the number of solutions of the equation $\sin^{-1} x=2 \tan^{-1} x$ is equal to

[2023 (13 Apr Shift 2)]

ANSWER KEYS

1. (4) **2.** (2)

1. (4)

Given,

$\sin^{-1}\left(\frac{x+1}{\sqrt{x^2+2x+2}}\right)-\sin^{-1}\left(\frac{x}{\sqrt{x^2+1}}\right)=\frac{\pi}{4}$

$\Rightarrow \sin^{-1}\left(\frac{x+1}{\sqrt{x^2+x+2}}\right)=\frac{\pi}{4}+\sin^{-1}\frac{x}{\sqrt{x^2+1}}$

$\Rightarrow\left(\frac{x+1}{\sqrt{x^2+x+2}}\right)=\sin\left(\frac{\pi}{4}+\sin^{-1}\frac{x}{\sqrt{x^2+1}}\right)$

$\Rightarrow\left(\frac{x+1}{\sqrt{x^2+x+2}}\right)=\frac{1}{\sqrt{2}}\times\frac{1}{\sqrt{x^2+1}}+\frac{1}{\sqrt{2}}\times\frac{x}{\sqrt{x^2+1}}$

$\Rightarrow \frac{x+1}{\sqrt{x^2+x+2}}=\frac{x+1}{\sqrt{2}\sqrt{x^2+1}}$

$\Rightarrow(x+1)\left(\sqrt{2}\sqrt{x^2+1}-\sqrt{x^2+x+2}\right)=0$

$\Rightarrow x=-1$ or $\sqrt{x^2+x+2}=\sqrt{2}\cdot\sqrt{x^2+1}$

Now solving, $\sqrt{x^2+x+2}=\sqrt{2}\cdot\sqrt{x^2+1}$ we get,

$x^2+x+2=2(x^2+1)$

$\Rightarrow x^2-x=0$

$\Rightarrow x=0,\ x=1$ {rejected as $x=1$ will not satisfy the given equation}

Hence, $S=\{0,1\}$

Now solving,

$\sum_{n\in S}\left(\sin(x^2+x+5)\frac{\pi}{2}-\cos(x^2+x+5)\pi\right)$

$=\left(\sin\frac{5\pi}{2}-\cos 5\pi+\sin\frac{5\pi}{2}-\cos 5\pi\right)$

$=1-(-1)+1-(-1)=4$

2. (2)

Given that $\sin^{-1}x=2\tan^{-1}x$

$\Rightarrow \sin^{-1}x=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

$\Rightarrow x=\left(\frac{2x}{1+x^2}\right)$

$\Rightarrow x+x^3=2x$

$\Rightarrow x^3-x=0$

$\Rightarrow x(x^2-1)=0$

$\Rightarrow x=0,-1,1.$

But $x\in(-1,1]$, so the number of roots are 2, $x=0$ & 1